

Week 5 Practical

To verify the Reader role, Sam Dlamini's account was used to attempt creating a new resource. Azure returned an 'Authorization Failed' error and the Create buttons were greyed out, confirming that the Reader role prevents any modifications to the environment.

Direction	Port / Protocol	Source / Dest	Action	Justification
Inbound	3389 / TCP	My IP	Allow	Allows RDP remote management of the VM from a trusted IP only, reducing attack surface
Inbound	443 / TCP	Any	Allow	Allows encrypted HTTPS web traffic to reach the hosted web application
Inbound	Any / Any	Any	Deny	Explicitly blocks all inbound traffic not covered by the above rules, enforcing least privilege
Outbound	Any / Any	Internet	Deny	Prevents data exfiltration by blocking VMs from sending data to the internet

LRS (Locally Redundant Storage) was selected because it replicates data three times within a single data centre, providing adequate redundancy for a small internal repository with no disaster recovery requirement. ZRS replicates across availability zones and GRS replicates to a secondary region — both are more expensive and unnecessary for this use case. The Hot access tier was chosen because the repository will be actively accessed by staff on a regular basis

SAS Token Details:

A read-only SAS token was generated for the public-assets container with an expiry window of 7 days (expiring 2026-05-21). A 7-day window was chosen because it provides sufficient time for authorised users to access the assets without leaving the token valid indefinitely. Read-only permissions were granted as the intended use is file retrieval only — no uploads or deletions are required.

SAS URL (redacted):

https://net731storage20231849.blob.core.windows.net/public-assets?sp=r&st=2026-05-15T14:53:47Z&se=2026-05-21T23:08:47Z&spr=https&sv=2025-11-05&sr=c&sig=EJMJ1hh0hJY4GfASKWmGqfy%2F78x4Xgfvuy10Nzv3KF[REDACTED]

Security Implications:

A SAS token grants access without requiring the caller to have an Azure AD identity, anyone with the URL can access the resource until it expires. If a SAS token was shared via email it could be intercepted, forwarded, or leaked. If a SAS token was accidentally exposed in a public GitHub repository, the immediate response would be to rotate the Storage Account keys, which instantly invalidates all SAS tokens generated from those keys. Future tokens should then be generated with shorter expiry windows and distributed through secure channels only.

Net731 | Access control (IAM)

Resource group

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Check accessRole assignmentsRolesDeny assignments

Number of role assignments for this subscription64000

Privileged4

View assignments

Search by name or email

Type: AllRole: AllScope: All scopesGroup by: Role

All (6)Job function roles (2)Privileged administrator roles (4)

Name	Type	Role	Scope	Condition
Owner (2)				
Jayden Hamilton41f144d1-ee9f-4796-91d2-ccf93ff7...	User	Owner	Subscription (Inherited)	None
Jayden Hamilton41f144d1-ee9f-4796-91d2-ccf93ff7...	User	Owner	Subscription (Inherited)	None
Contributor (2)				
Jayden Hamiltonbd1b063b-9188-4424-b8c3-db730...	User	Contributor	This resource	None

Contributor (1)

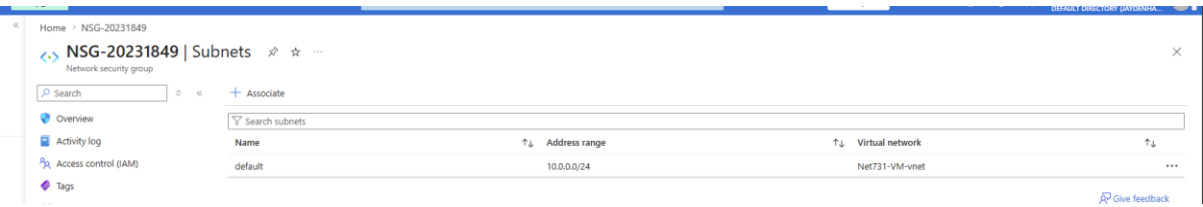
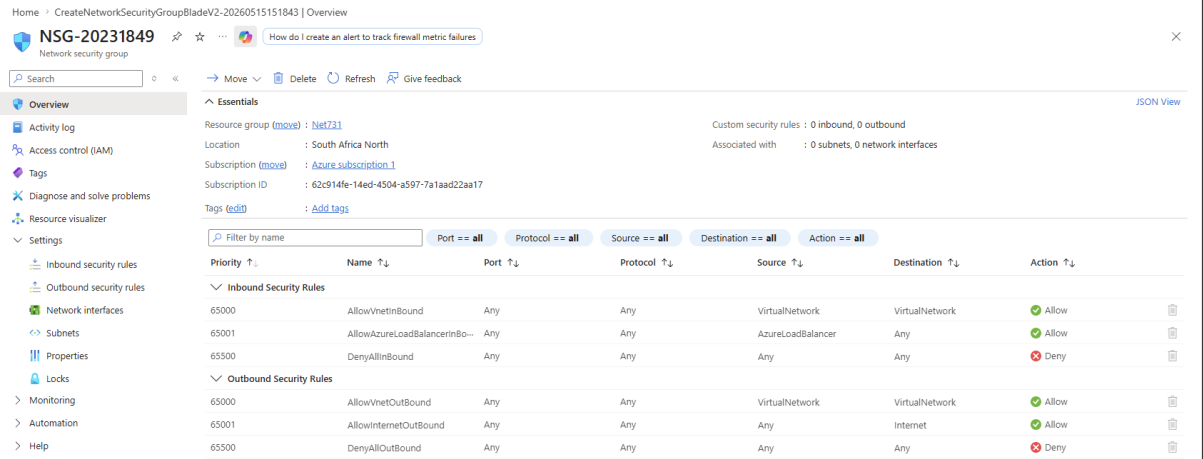
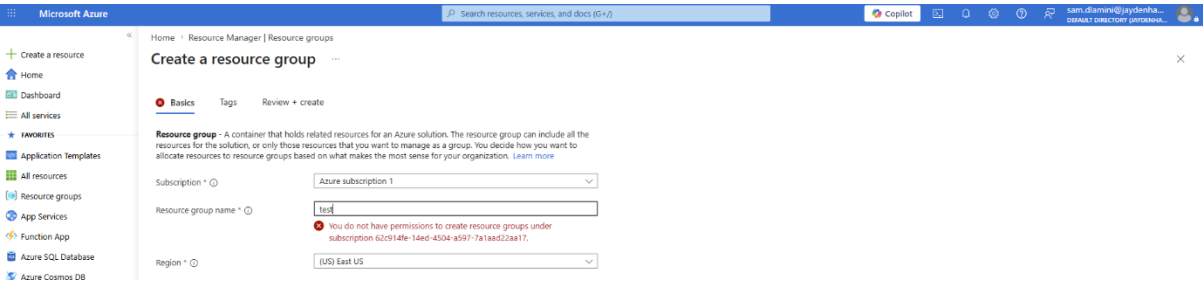
Jayden Hamiltonbd1b063b-9188-4424-b8c3-db730...	User	Contributor	This resource	None
---	------	-------------	---------------	------

Reader (1)

Sam Dlamini f226d6c5-9c65-45fb-8f46-8f5ee97c...	User	Reader	This resource	None
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Storage Blob Data Contributor (1)

Morgan Sithole0da011cf-a297-4b15-8b79-6b70e8...	User	Storage Blob Data Contributor	This resource	Add
---	------	-------------------------------	---------------	-----



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net731storage20231849

Storage account

Help me reduce the costs of my storage account

Does this storage account follow security best practices

How can I make my storage account more resilient?

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Delete

Move

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Open in mobile

CU / PS

Feedback

Essentials

Resource group (move) : Net731

Location : southafricanorth

Subscription (move) : Azure subscription 1

Subscription ID : 62c914fe-14ed-4504-a597-7a1aad22aa17

Disk state : Available

Tags (edit) : Add tags

Performance : Standard

Replication : Locally-redundant storage (LRS)

Account kind : StorageV2 (general purpose v2)

Provisioning state : Succeeded

Created : 2026/05/15, 16:54:44

Properties

Monitoring

Capabilities (7)

Recommendations (0)

Tutorials

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Blob service

Hierarchical namespace : Disabled

Default access tier : Hot

Blob anonymous access : Disabled

Blob soft delete : Enabled (7 days)

Container soft delete : Enabled (7 days)

Versioning : Disabled

Change feed : Disabled

NFS v3 : Disabled

Allow cross-tenant replication : Disabled

Storage tasks assignments : None

File service

Identity-based access : Not configured

Default share-level permissions : Disabled

Soft delete : Enabled (7 days)

Require Encryption in Transit for SMB (Security) : Enabled

Security

Require secure transfer for REST API operations : Enabled

Storage account key access : Enabled

Minimum TLS version : Version 1.2

Infrastructure encryption : Disabled

Networking

Public network access : Enabled

Public network access scope : Enable from all networks

Private endpoint connections : 0

Network routing : Microsoft network routing

Endpoint type : Standard

Add or remove favorites by pressing Ctrl+Shift+F

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net731storage20231849

Containers

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Delete

Change access level

Restore containers

Edit columns

Search containers by prefix

Only show active containers

Showing all 3 items

☐	Name	Last modified	Anonymous access level	Lease state	Storage connector
☐	logs	2026/05/15, 16:55:10	Private	Available	-
☐	internal-docs	2026/05/15, 17:00:41	Private	Available	-
☐	public-assets	2026/05/15, 17:01:03	Private	Available	-

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internal-docs

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Acquire lease

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internal-docs

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	Media.jpg	2026/05/15, 17:04:02	Hot (Inferred)	Block blob	82.87 KiB	Available

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public-assets

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+ Add Directory

Upload

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Rename

Acquire lease

Break lease

Edit columns

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public-assets

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	Media.jpg	2026/05/15, 17:08:08	Hot (Inferred)	Block blob	82.87 KiB	Available

Successfully uploaded blob(s)

Successfully uploaded 1 blob(s).

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public-assets | Shared access tokens

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Metadata

A shared access signature (SAS) is a URI that grants restricted access to an Azure Storage container. Use it when you want to grant access to storage account resources for a specific time range without sharing your storage account key. [Learn more about creating an account SAS](#)

Signing method

Account key

User delegation key

Signing key

Key 1

Stored access policy

None

Permissions

Read

Start and expiry date/time

Start

05/15/2026

4:53:47 PM

(UTC+02:00) Harare, Pretoria

Expiry

05/22/2026

1:08:47 AM

(UTC+02:00) Harare, Pretoria

Allowed IP addresses

for example, 168.1.5.65 or 168.1.5.65-168.1...

Allowed protocols

HTTPS only

HTTPS and HTTP

Generate SAS token and URL

Blob SAS token

sp=r&st=2026-05-15T14:53:47Z&se=2026-05-21T23:08:47Z&spr=https&sv=2025-11-05&sr=c&sig=EIMJ1hh0hY4GfASKWmGqfy%2F78x4Xgivyuy10Nzv3Kfg%3D

Blob SAS URL

https://net731storage20231849.blob.core.windows.net/public-assets?sp=r&st=2026-05-15T14:53:47Z&se=2026-05-21T23:08:47Z&spr=https&sv=2025-11-05&sr=c&sig=EIMJ1hh0hY4GfASKWmGqfy%2F78x4Xgivyuy10Nzv3Kfg%3D

Add or remove borders by pressing Ctrl+Shift+F

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☐	internal-docs	2026/05/15, 17:11:29	Private	Available	-
☐	public-assets	2026/05/15, 17:01:03	Private	Available	-